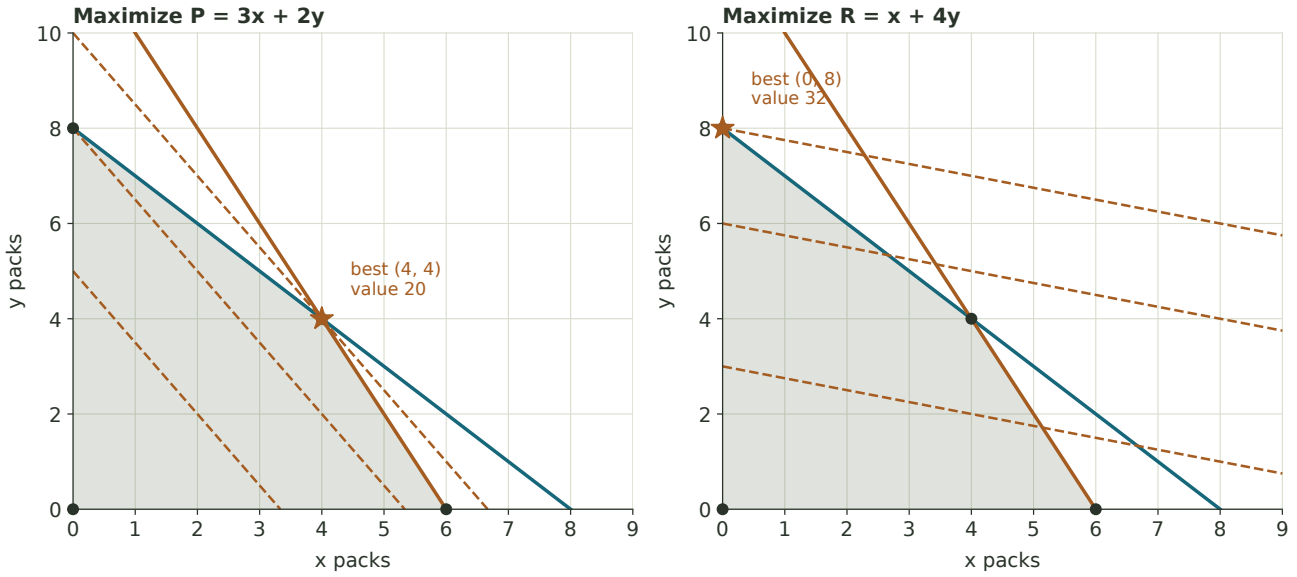
WEEK 8 / PART II / LINEAR PROGRAMMING

# Which plan is best?

Keep the allowable choices fixed, state a goal, and compare the consequences. A mathematical optimum answers the question you actually put into the model.




**From geometry to a decision**

The club's feasible region has corners $(0,0), (6,0), (4,4), (0,8)$. With contribution $P = 3x + 2y$, four packs of each type give the largest total. A different goal can choose a different corner, or leave an entire edge tied.

A feasible point represents an allowable plan. A linear objective assigns a score to each plan; examining the feasible region identifies the highest or lowest attainable score, including ties and possible unboundedness.

For the core method you need only substitution and comparison. Equal-score lines and convex mixtures explain why the method works. The whole-number investigation is an extension.

Online reading · Optimizer · Optimization extension

## 8.6 An objective gives each plan a score

Suppose the shop earns $30 after replacement parts for each tune-up and $20 for each quick repair. The total contribution for a day is

$$P = 30x + 20y.$$

This is the *objective function*: a rule assigning a number to each plan. The letter P names the total contribution in dollars. To *maximize* it means to find the largest value among feasible plans.

**Example · Feasible does not mean optimal**

Three tune-ups and three quick repairs give $P = 30(3) + 20(3) = 150$ dollars. Four of each give $P = 30(4) + 20(4) = 200$ dollars. Both are possible schedules, but the second earns more.
Five tune-ups and three quick repairs would bring in $210. A larger earnings figure cannot make an impossible schedule workable: it requires more mechanic time than the shop has.

A function such as $30x + 20y$ is *linear*: every additional tune-up adds the same $30 and every quick repair adds the same $20. There is no product $xy$, squared quantity, or changing rate. A *linear program* combines a linear objective with linear constraints. Here "programming" means planning; writing computer code is not required.

**Pause and try 1**

At (2,5), what is P=30x+20y? Does knowing the money earned tell you whether the schedule is feasible?

## 8.7 For this bounded polygon, compare the corners

Our continuous feasible region contains infinitely many points. The *corner-point principle* says that a linear objective on a nonempty, closed, bounded polygonal feasible region attains its maximum and minimum at vertices. "Bounded" means it fits inside some finite box. "Closed" here means the boundary is included.

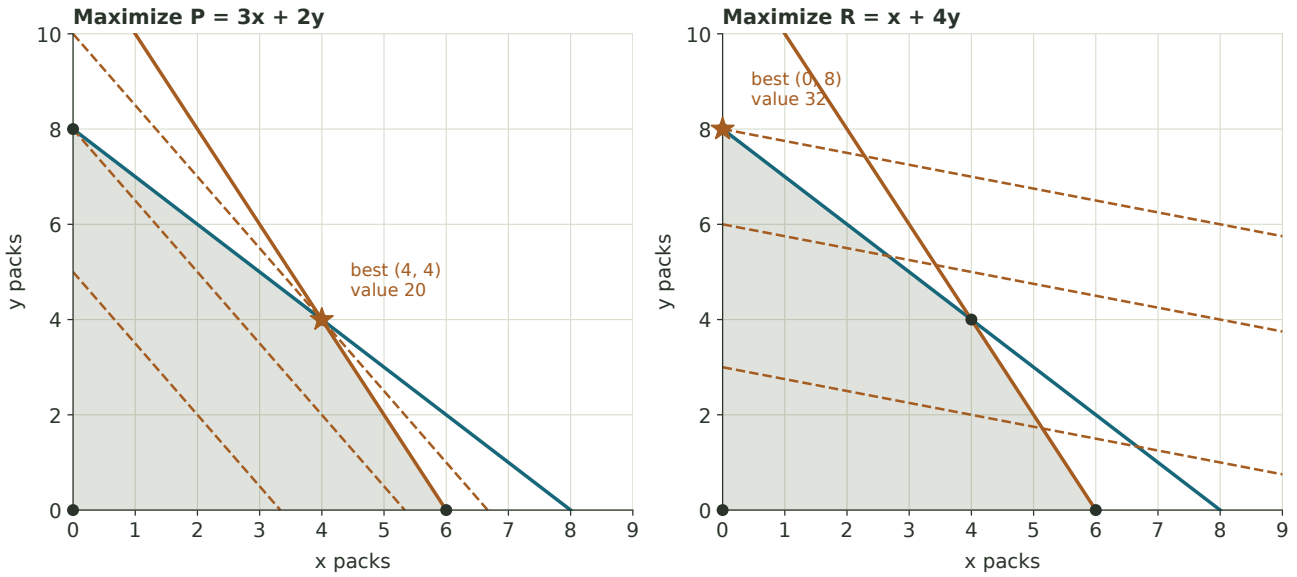
This guarantees that at least one vertex gives the best value. It does not guarantee a unique best point, or say that every best point must be a vertex. Those distinctions matter when there is a tie.

| Feasible vertex | Calculation of P = 30x + 20y | Contribution |
|---|---|---|
| (0,0) | 30(0)+20(0) | $0 |
| (6,0) | 30(6)+20(0) | $180 |
| (4,4) | 30(4)+20(4) | $200 |
| (0,8) | 30(0)+20(8) | $160 |

The largest contribution is $200 at (4,4). The shop should schedule four tune-ups and four quick repairs. Report both the schedule and the money earned; they answer different questions.

**Why a corner comparison can work**

Any point in a filled convex polygon can be expressed as a mixture of its vertices. A linear objective at that point is the same mixture of the vertex scores. An average of scores cannot exceed the largest score or fall below the smallest. This is the geometric reason the table can locate an optimum.




Each dashed line joins plans with an equal objective value. Moving to a parallel line changes that score, not the resource constraints. The star marks the last feasible contact in the direction of increasing score.

In our model, the best plan is also a whole-number plan, so it remains optimal if we restrict production to whole bicycles. A continuous optimum is an upper bound for the integer choices; when a whole-number plan attains it, the integer problem is solved too.

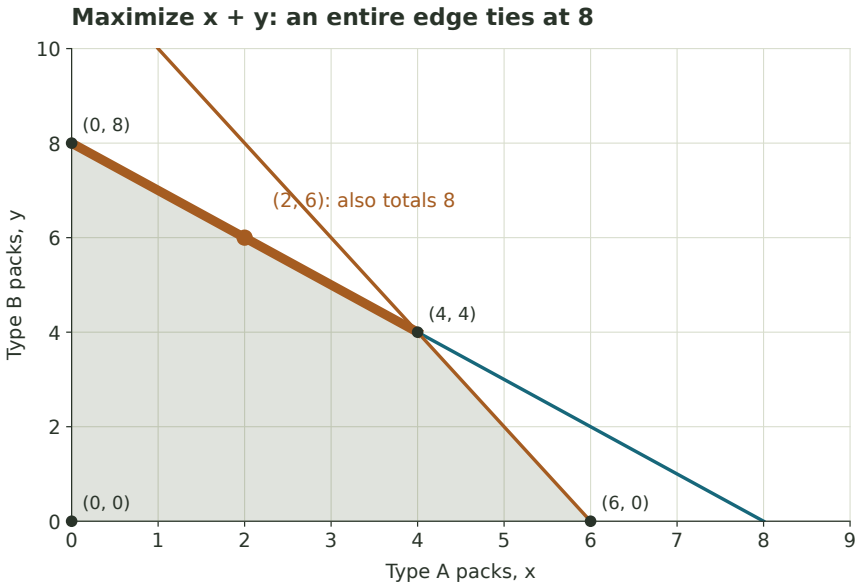
**Pause and try 2**

Why is checking just the two axis corners insufficient?

## 8.8 The feasible region stays; the goal can change

Suppose a new service contract favors quick repairs. To see how a change in priorities changes the answer, use the alternative score $R = x + 4y$. At the same four vertices its values are 0, 6, 20, and 32. Now (0,8) is best. The resource limits and region did not change. The objective changed.

Alternatively, suppose the shop wants to return the largest *number of bikes* to customers, regardless of type. Maximize $T = x + y$. The eight-bike limit already tells us $T \leq 8$. Both (4,4) and (0,8) attain 8, and every point on the segment between them also attains 8.




These are multiple optimal solutions. If actual bicycles must be whole, the five plans (0,8), (1,7), (2,6), (3,5), and (4,4) all attain the same largest total.

### Example · A tie leaves room for another consideration

If the goal is to help as many customers as possible, (4,4) and (0,8) both serve eight bikes. But the first needs twelve mechanic-hours and the second needs only eight. To save mechanic time, choose (0,8); to earn more after parts, choose (4,4).

A second goal can distinguish plans that tie under the first. We should name that goal rather than pretend the first objective made the whole decision.

### Pause and try 3

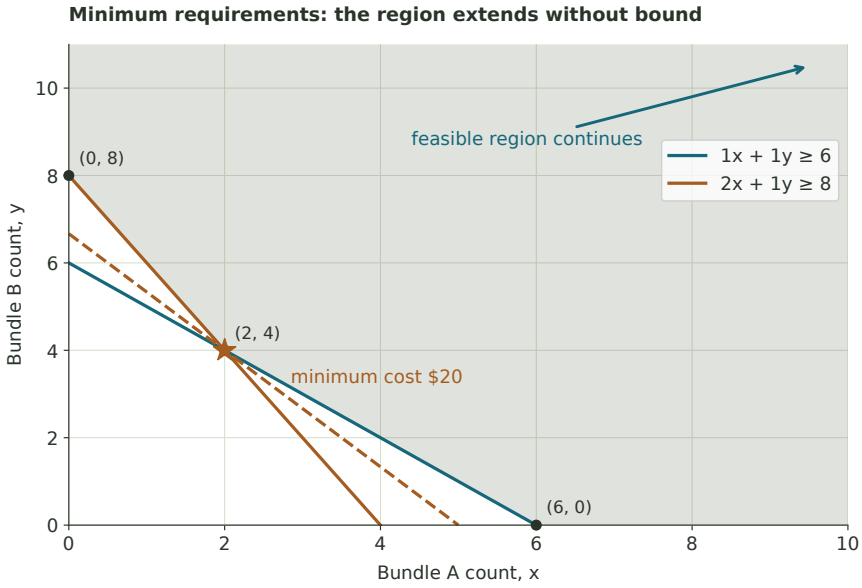
If two adjacent feasible vertices both give the maximum score for a linear objective, what happens between them? Does the same conclusion follow merely because two vertices have equal scores?

## 8.9 "At least" often reverses the feasible side

A neighborhood group is organizing a back-to-school supply drive. A store offers two bundles. Bundle A contains 1 notebook and 2 pencils and costs \$4. Bundle B contains 1 notebook and 1 pencil and costs \$3. The group needs at least six notebooks and eight pencils for the students it is helping. Let x and y be the respective bundle counts.

$$x + y \geq 6, \qquad 2x + y \geq 8, \qquad x, y \geq 0; \qquad \text{minimize } C = 4x + 3y.$$

We can buy extra supplies, so there is no upper limit in this model. The feasible region extends up and to the right. It is *unbounded*, but that does not mean its minimum cost is missing.




The top and right edges of the picture are viewing limits, not extra constraints. The arrow shows the feasible region continuing. Buying more and more bundles raises this positive cost; it does not lower it indefinitely.

| Boundary vertex | Notebooks | Pencils | Cost 4x+3y |
|---|---|---|---|
| (0,8) | 8 | 8 | \$24 |
| (2,4) | 6 | 8 | \$20 |
| (6,0) | 6 | 12 | \$24 |

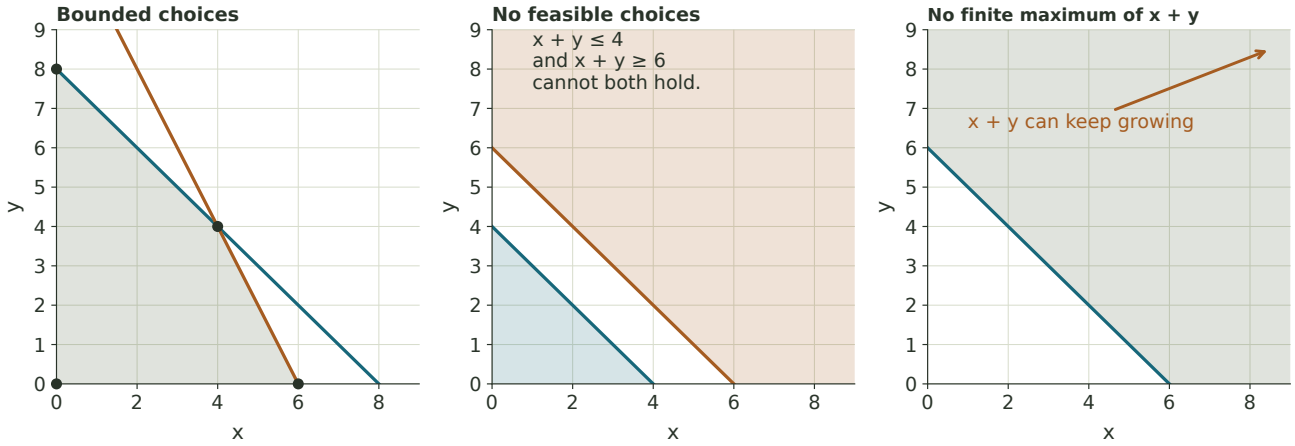
The group can meet both requirements for \$20 with two A-bundles and four B-bundles. For this nonnegative-variable model, the least cost occurs at a feasible vertex. On an unbounded region, we must also check whether some allowed direction keeps improving the objective. Here all costs are positive, so going farther out cannot make the cost decrease forever.

Buying nothing costs zero but would leave students without the supplies. This is why minimizing cost cannot be separated from stating what must be provided. If the only restrictions were upper limits and nonnegativity, doing nothing could indeed be the mathematical minimum-cost plan.

**Pause and try 4**

Why is (0,6) not feasible for the supply order even though it provides all six notebooks?

## 8.10 A model can have a best plan, no plan, or no finite best value




Separate the questions: Is any plan feasible? Is the region bounded? Can the objective keep improving without limit? An unbounded region can still have a finite minimum, as the supply-order example showed.

- *Infeasible model:* no point satisfies every constraint. For example, $x + y \leq 4$ and $x + y \geq 6$ cannot both hold. There is no permissible plan to optimize.
- *Unbounded objective:* feasible plans exist, but their scores improve without limit. For example, maximizing $x + y$ with only $x + y \geq 6$ and nonnegativity allows arbitrarily large totals.
- *Finite optimum:* some feasible plan attains the best value. It may be one point, an edge, or a larger set when scores tie.

The optimizer identifies these different outcomes. It computes vertices from the actual constraints; a corner of its graph window is not automatically a real feasible vertex.

**Included boundaries matter too**

With $0 \leq x < 1$, there is no largest allowable x: any proposed x can be increased a little while staying below 1. The value 1 is a limiting bound, but it is excluded. Our corner method uses non-strict inequalities so the relevant boundaries are available.

**Pause and try 5**

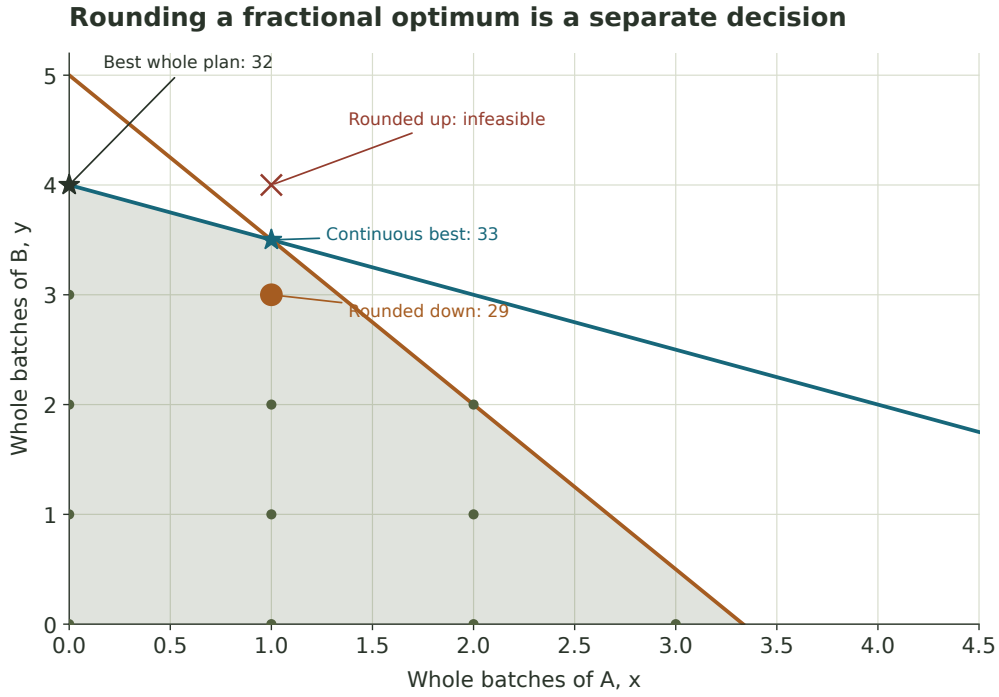
Does an unbounded feasible region always imply that there is no minimum? Give an example from this reading.

## Extension: whole-number plans

Acres, mixture amounts, and hours can often be treated as divisible. Chairs, buses, and repaired bicycles usually cannot. Requiring whole-number decisions adds an *integer constraint*. It changes which points are allowed, even when the drawn continuous region stays the same.

### Example · Rounding can fail in two ways

Maximize $S = 5x + 8y$ subject to $x + 2y \leq 8$, $3x + 2y \leq 10$, and nonnegativity. The continuous best point is (1,3.5), with score $5(1) + 8(3.5) = 33$.
Rounding up to (1,4) violates both limits: the totals become 9 and 11. Rounding down to (1,3) is feasible but scores only 29. The whole-number plan (0,4) is feasible and scores 32, so rounding down missed a better plan.




For a small model, list feasible whole-number choices systematically. A fractional continuous optimum bounds the possible whole-number score from above, but it may not itself be usable.

To verify the integer optimum here, try y = 0, 1, 2, 3, 4. The largest allowable whole x-values are 3, 2, 2, 1, 0, giving respective scores 15, 18, 26, 29, 32. This finite check establishes the best whole plan.

### Pause and try 6

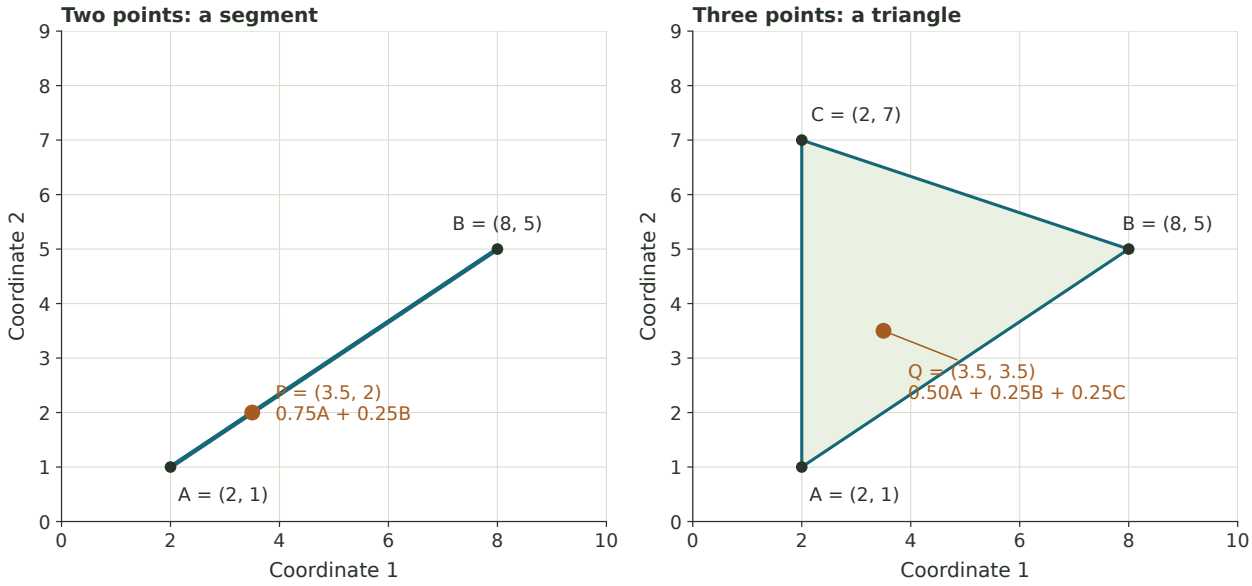
What does the continuous maximum 33 establish about any feasible whole-number score in this example? Why does it not establish that 33 can be achieved with whole numbers?

## 8.11 A convex combination is a weighted average of points

Weeks 2–3 used fractions, relative weights, and averages to summarize numbers. We can average points by averaging their coordinates. For points $A = (2, 1)$ and $B = (8, 5)$, use weights 0.75 and 0.25:

$$P = 0.75A + 0.25B = (0.75(2) + 0.25(8),\ 0.75(1) + 0.25(5)) = (3.5, 2).$$

A *convex combination* uses nonnegative weights that add to 1. For two points, all these combinations fill the segment between them. For three noncollinear points, they fill the triangle. The smallest convex set containing the given points is their *convex hull*.




A weight of 0.75 on A places the result nearer A, one quarter of the way from A toward B. The numbers are coordinates in an invented representation; their axes have no assumed physical units.

The AI application isolates one use of this idea: a normalized nonnegative weighted combination of value vectors in attention. The geometric guarantee concerns that one averaging step. Later transformations in a model need not stay in the original hull, and the weights do not certify that an interpretation is true.

**Pause and try 7**

Why do weights 0.6 and 0.4 define a convex combination, while 1.2 and $-0.2$ do not?

## 8.12 A clear recommendation names its goal and assumptions

Linear models assume constant rates: each tune-up takes the same two hours and earns the same \$30 after parts; each quick repair takes one hour and earns \$20. Actual work can involve cancellations, unusually difficult repairs, different mechanics, and changing parts prices. The graph describes the consequences of the assumptions we have made.

Increasing a resource allowance in a "$\leq$" constraint expands the feasible set or leaves it unchanged. Therefore the best maximum score cannot decrease. But it need not improve: another resource may still block every better plan. Tightening a limit can shrink the set or even make it empty.

**Example · More mechanic-hours do not create more customers**

Keep $x + y \leq 8$, but suppose the shop can now staff sixteen mechanic-hours. It could schedule eight full tune-ups: (8,0) involves eight bikes and sixteen hours and earns \$240 after parts. Adding more hours alone would not increase the maximum while the shop still accepts only eight bikes. At \$30 per tune-up, eight bikes could earn at most \$240.

## A practical routine

1. Identify what the people involved can choose, what the units mean, and whether fractions make sense.
2. Translate every resource limit and minimum requirement into a constraint.
3. State the objective and whether to maximize or minimize it.
4. Find all feasible vertices and check candidate intersections.
5. Compare objective values and identify ties or improving directions.
6. Check whole-number restrictions and interpret the plan and value.
7. Ask whom the decision affects, what the model leaves out, and what information could change the recommendation.

Week 9 adds another reason a plan might be impossible: locations and resources must be connected by usable routes. Keep the same distinction there – what is allowed, what is preferred, and what the model leaves out.

# Further Worked Models

## Furniture: compare every feasible corner

A furniture maker produces chairs $x$ and desks $y$ per week. Two resource constraints are $2x + 3y \leq 120$ and $x + 2y \leq 70$, with nonnegativity. Maximize contribution $P = 40x + 70y$.

The first boundary has intercepts $(60, 0)$ and $(0, 40)$; the second has $(70, 0)$ and $(0, 35)$. The tighter axis limits give $(60, 0)$ and $(0, 35)$. To find the non-axis intersection, double $x + 2y = 70$ and subtract $2x + 3y = 120$: $y = 20$, then $x = 30$.

| Corner | First resource | Second resource | Contribution |
|---|---|---|---|
| $(0, 0)$ | 0 | 0 | \$0 |
| $(60, 0)$ | 120 | 60 | \$2,400 |
| $(30, 20)$ | 120 | 70 | \$2,600 |
| $(0, 35)$ | 105 | 70 | \$2,450 |

All four corners pass both limits. The maximum is \$2,600 at 30 chairs and 20 desks. The plan and its value are different quantities; both belong in the recommendation.

## A resource limit can be redundant

Suppose a different model has limits $x + 2y \leq 20$ and $x + 2y \leq 24$. Every point passing the first also passes the second. The second is redundant. This differs from merely having slack at one plan: a constraint can be slack at a particular optimum yet still exclude other choices elsewhere.

## A third real constraint can create a fifth corner

A print shop has three limits:

$$x + 2y \leq 60, \quad x + y \leq 40, \quad 2x + y \leq 70, \quad x, y \geq 0.$$

Here $x$ counts posters and $y$ banners. Maximize the invented contribution $P = 15x + 35y$.

Intersect the first two non-axis boundaries to get $(20, 20)$; the third total is 60, so it passes. Intersect the second and third to get $(30, 10)$; the first total is 50, so it passes. The first and third meet at $(80/3, 50/3)$, but this candidate has $x + y = 130/3 > 40$, so it must be discarded.

The x-axis corner is $(35, 0)$: the third constraint is tighter there than the other two. The y-axis corner is $(0, 30)$: the first constraint is tighter there. Together with the origin and the two valid non-axis intersections, these give five corners.

| Feasible corner | Calculation | Contribution |
|---|---|---|
| $(0, 0)$ | $15(0) + 35(0)$ | \$0 |
| $(35, 0)$ | $15(35) + 35(0)$ | \$525 |
| $(30, 10)$ | $15(30) + 35(10)$ | \$800 |
| $(20, 20)$ | $15(20) + 35(20)$ | \$1,000 |
| $(0, 30)$ | $15(0) + 35(30)$ | \$1,050 |

The maximum is \$1,050 at $(0, 30)$. All three resource constraints matter to this feasible region; none is redundant. A calculation at a visually prominent intersection, such as $(20, 20)$, is not a substitute for comparing all corners.

### A useful distinction

A constraint is *binding at a plan* when it holds with equality there. It is *redundant in a model* when removing it leaves the entire feasible set unchanged. These are different claims.

## Part II Review

Use nonnegative variables and included boundaries. Show a corner table or explicit calculation and report both a plan and its objective value.

1. For the club's corners $(0,0), (6,0), (4,4), (0,8)$, maximize $P = 3x + 2y$. What is the minimum if there is no lower production requirement?
2. On the same region, maximize $R = x + 4y$ and then $T = x + y$. Describe the complete optimal edge for $T$, including its whole-number points.
3. Maximize $f = 5x + 8y$ subject to $x + 2y \leq 16$, $3x + 2y \leq 24$. Find all corners and compare. Check resource use at the best plan.
4. Minimize $C = 3x + 5y$ subject to $x + y \geq 8$, $2x + y \geq 10$. Find the boundary vertices. Explain why the mixed boundary intersection is not automatically best.
5. For the furniture model in the worked example, compare the best plan with producing only desks. Does the higher per-item contribution determine the answer by itself?
6. For the three-limit print shop, why is $(80/3, 50/3)$ not a feasible corner? Which corner maximizes $15x + 35y$?
7. A toy workshop has $3x + y \leq 120$, $x + 2y \leq 80$, and contribution $P = 8x + 5y$. Find the corners and best plan. Then change the contribution to $5x + 8y$ and compare.
8. A municipality divides an invented \$500-thousand budget between roads $x$ and parks $y$, measured in thousands of dollars. Require $x \geq 100$, $y \geq 80$, $x + y \leq 500$. Maximize an invented score $S = 3x + 4y$. What does the score leave for people to decide?
9. A polygon has corners $(0,0), (8,0), (5,4), (0,6)$. For cost $C = 4x + 6y$, which of the three listed nonzero corners is cheapest? Why is that not the minimum over the full polygon? Give a different positive feasible plan with lower cost.
10. For $x + 2y \leq 8$, $3x + 2y \leq 10$, maximize $5x + 8y$ continuously and then using whole numbers. Check rounding up and rounding down from the continuous best point.
11. Compute $0.75A + 0.25B$ for $A = (2,1), B = (8,5)$. Add $C = (2,7)$ and normalize amounts $2, 1, 1$. Where do the resulting points lie? Why do weights $1.2, -0.2$ fail the convex-combination test?
12. Explain the difference between an infeasible model and an unbounded objective. Give an example of an unbounded region with a finite minimum. Explain why $0 \leq x < 1$ has no largest allowable x.
13. Keep $x + y \leq 8$ and increase the club's ingredient allowance from 12 to 16, then 20. Maximize $3x + 2y$ at each stage. Why does the extra ingredient allowance eventually stop helping?
14. Recommend a plan for the original chair-and-table workshop with wood limit 40, assembly limit 14, and contribution $30x + 50y$. Use resource totals and a comparison with another feasible plan. Name a relevant missing feature.

## Solutions and Reasoning

**Short check 1.** P=30(2)+20(5)=160 dollars. The score alone does not prove feasibility; check the constraints. Here the schedule has 7 bikes and uses 9 mechanic-hours, so the plan is feasible.

**Short check 2.** The best corner can occur where two non-axis boundaries meet. Here (4,4) gives \$200, exceeding the \$180 and \$160 at the axis corners. The method requires every feasible vertex.

**Short check 3.** Every point on the edge joining two optimal adjacent vertices has the same maximum score. Equal scores alone do not establish optimality: some other vertex might score higher. First compare all feasible corners.

**Short check 4.** It supplies only 6 pencils, below the minimum 8. Both requirements must hold. A boundary intercept of one requirement may fail another.

**Short check 5.** No. The supply-order region extends indefinitely, but positive costs give a minimum of \$20 at (2,4). Unbounded choices and an unbounded objective are different properties.

**Short check 6.** Every whole-number feasible plan is also continuously feasible, so its score is at most 33. But the point giving 33 has a fractional coordinate, so it does not establish an attainable whole-number score.

**Short check 7.** The first pair is nonnegative and sums to 1. The second also sums to 1 but has a negative weight; it can extrapolate beyond the segment. Both conditions are needed.

### Part II Review: solutions

1. Corner values are $0, 18, 20, 16$. Maximum \$20 is at $(4, 4)$. With only upper limits and nonnegativity, minimum contribution is 0 at zero production. A minimum-cost model may need minimum service requirements.

2. R-values are $0, 6, 20, 32$, so $(0, 8)$ is best. T reaches 8 along the edge from $(4, 4)$ to $(0, 8)$. Its whole-number plans are $(0, 8), (1, 7), (2, 6), (3, 5), (4, 4)$.

3. Subtraction gives $x = 4, y = 6$. Corners are $(0, 0), (8, 0), (4, 6), (0, 8)$; scores are $0, 40, 68, 64$. The maximum is 68 at $(4, 6)$, using 16 units of the first resource and 24 of the second.

4. Corners are $(8, 0), (2, 6), (0, 10)$, with costs $24, 36, 50$. The minimum is 24 at $(8, 0)$. It provides 8 of the first requirement and 16 of the second, exceeding its 10-unit minimum by 6. A feasible axis corner can outperform a mixed intersection.

5. $(30, 20)$ gives \$2,600; desks only gives $(0, 35)$ and \$2,450. The best mix depends on both resource limits, not just the larger contribution per item.

6. That intersection has $x + y = 130/3 > 40$ and violates the middle constraint. The maximum is \$1,050 at $(0, 30)$, which passes all three limits.

7. Corners are $(0, 0), (40, 0), (32, 24), (0, 40)$. Original scores are $0, 320, 376, 200$, so $(32, 24)$ is best at 376. Under $5x + 8y$, scores are $0, 200, 352, 320$; the same plan remains best but its score changes. Changing an objective need not always move the optimum.

8. Corners are $(100, 80), (420, 80), (100, 400)$. Scores are $620, 1580, 1900$. The score selects roads \$100,000 and parks \$400,000. Its coefficients and minimum allocations are modeling choices, not facts establishing what a community ought to value.

9. The three costs are 32,44,36, so $(8, 0)$ is cheapest among those candidates. The full polygon includes $(0, 0)$ at cost 0 and, for example, $(1, 0)$ at cost 4. A real production floor needs an explicit constraint, not deletion of one point.

10. Continuous maximum 33 is at $(1, 3.5)$. Rounding up to $(1, 4)$ gives resource totals 9 and 11, both too large. Rounding down to $(1, 3)$ is feasible but scores 29. The best whole plan is $(0, 4)$ at 32; enumerate y-values 0–4 to verify.

11. The first point is $(3.5, 2)$, on the A–B segment. Normalized amounts give weights $(0.5, 0.25, 0.25)$ and point $(3.5, 3.5)$ inside the triangle. A negative weight violates nonnegativity even if the weights sum to 1.

12. Infeasible means no plan satisfies every condition. Unbounded objective means feasible plans improve the score without finite limit. The supply-order model $x + y \geq 6$, $2x + y \geq 8$ has unbounded choices but minimum $4x + 3y = 20$ at $(2, 4)$. Under $0 \leq x < 1$, any allowed x can be increased while remaining below 1; the limiting bound 1 is excluded.

13. With allowance 12, the maximum is 20 at $(4, 4)$. At 16 and 20, it is 24 at $(8, 0)$. The packing limit permits at most eight packs, and each contributes at most 3; no plan can exceed 24 while that limit stays fixed. Beyond 16, the ingredient constraint is redundant.

14. Produce 8 chairs and 6 tables: wood use $2(8) + 4(6) = 40$, assembly use $8 + 6 = 14$, contribution \$540. Ten tables alone are feasible but contribute \$500. Demand, waste, setup time, resource uncertainty, and variable production rates are examples of missing features that could change a real recommendation.

## Optional primary source for the AI connection

Vaswani et al., *Attention Is All You Need*, Section 3.2. The geometry applies to the normalized weighted value combination at that step; later transformations can move a representation beyond the original hull. The vectors and weights in this unit are constructed examples.